

Optimal Correction of Environmental Stochastic Externalities

Raphaël Jananji* and Alexandre Pavlov†

April 15, 2026

Work in Progress: **Latest version**

Abstract

How does pollution affect climate risk? The scientific literature views this question in two main ways. First, the "tipping point" view according to which pollution increases the probability of discrete catastrophic events. Second, the "distributional shift" view according to which pollution shifts the entire distribution of climate shocks and makes extreme weather events more frequent and more intense. In this paper, we study the implications of the second type of risk on the social cost of carbon (SCC), so far unexplored in the economic literature. We model what we call stochastic externalities in a framework with random climate shocks where the parameters of its distribution are functions of the pollution stock rather than fixed values. We solve the social planner's problem in a representative agent economy in a simple three-period model which we then generalize to an infinite-horizon recursive framework. The modified SCC sums a stochastic externality correction term to the standard Pigouvian term, which may be expressed in different ways but is essentially an expectation of future damages weighted by the distribution's sensitivity to climate variables. We provide a few illustrative examples: for Pareto shocks, the correction has a single tail-fattening channel which is an expectation of future damages weighted by an exponentially-growing term. This implies a stochastic externality correcting term which has the potential to grow faster without bound than the standard Pigouvian term, reinforcing Weitzman's concerns of a Dismal Theorem. In a simple numerical example, we show how the stochastic externality may significantly increase the SCC beyond the standard Pigouvian level. Our framework allows for any parametric family of continuous distributions satisfying mild regularity conditions and nests the standard SCC as a special case.

Keywords: *Optimal environmental taxation; stochastic externalities; carbon tax; endogenous climate risk; social cost of carbon; Pigouvian taxation*

JEL Classification: *D62; D82; H21; H23*

*raphael.jananji@umontreal.ca

†alexandre.pavlov@umontreal.ca

1 Introduction

How does pollution affect climate risk? The scientific literature views this question in two main ways. First, the "tipping point" view according to which pollution increases the probability of discrete catastrophic events. Second, the "distributional shift" view according to which pollution shifts the entire distribution of climate shocks and makes extreme weather events more frequent and more intense. These two views are different yet complementary: one is concerned with major regime shifts in the climate system that may happen in the future, while the other is concerned with the continuous increase of "everyday" climate risk, such as extreme events getting stronger and less predictable. Although a "tipping point" may be a civilizational threat, or indeed even an existential threat, the "distributional shift" perspective is nevertheless worth studying since the economy is already arguably significantly affected to this type of risk and will continue suffer from it in the future. Indeed, according to the NOAA, the number of billion-dollar weather disasters in the US has been increasing over the past decades, going from an average of 8.5 between 1980 and 2023 to 20.4 between 2017 and 2023 (NCEI, 2024). More recently, Bakkensen and Barrage (2026) empirically estimates the effect of climate change on future cyclone risk in the United States and the Caribbean using NOAA simulation data, finding that climate change will cause an almost threefold increase in welfare losses from cyclones in the United States.

In this paper, we develop a framework to analyze the implications of a "distributional shift" on the social cost of carbon (SCC). We model what we call stochastic externalities in a framework with random climate shocks where the parameters of its distribution are functions of the pollution stock rather than fixed values, as is common in DSGE economic modeling. First, to build intuition, we enhance a standard three-period neoclassical growth model where current emissions harm a fraction of future output and derive a SCC which sums a standard Pigouvian correction term and a stochastic externality correction term. There are many ways to express this second term but the conceptually simplest form may be the quantile form, where the expectation of future marginal damages is weighted by the change in the quantile function of the shock distribution caused by the increase in the pollution stock. In other words, marginal utility of damages is weighted by how much more or less likely a particular shock realization becomes after the pollution-induced distributional shift.

We then provide a few illustrative examples to show how this translates into more concrete economically-meaningful channels. In the normal distribution case, the stochastic externality correction decomposes into three channels: a mean-shift channel, a risk-aversion channel, and a damage-curvature channel. In the Pareto distribution case, the stochastic externality correction has a single tail-fattening channel which is an expectation of future damages weighted by an exponentially-growing term. This implies a stochastic externality correcting term which has the potential to grow faster without bound than the standard Pigouvian term, reinforcing Weitzman's concerns of a Dismal Theorem. In a simple numerical example, we show how the stochastic externality may significantly increase the SCC beyond the standard Pigouvian level. However, our framework also allows for any parametric family of continuous distributions satisfying mild regularity conditions and nests the standard SCC as a special case, when shocks are "exogenous".

Afterwards, to bridge theory and practice, we generalize this framework to a recursive infinite

horizon model. Although the standard recursive framework invokes Markov properties to argue for the validity of a recursive form (see for example Stokey et al., 1989), we can still involve Markov *Decision* Process arguments. Decision Markov processes are a generalization of Markov processes where the past state affects not only the current state, but also the probability distributions. Once the foundations are laid out, we show how this new "stochastic externality" channel affects the SCC in practice, modifying the DICE benchmark model to include a simple endogenous distributional shift effect. This example is illustrative and requires more realistic calibration, but nevertheless shows the practical applicability of this framework. Moreover, we use the normal distribution, which will by design imply a lower effect of stochastic externalities on the SCC, since it is thin-tailed and symmetrical. Further work will use more realistic calibrated thick-tailed distributions, drawing from both the scientific literature and the emerging microeconomic literature on the effects of weather extremes.

Before moving to the model, let us quickly summarize the current state of the literature to better position this work. This paper is not the first to model "endogenous" climate risk as there already exists a wealth of literature on the topic. For example, papers like Barro (2015), Brausmann and Bretschger (2024), Cai and Lontzek (2019), Müller-Fürstenberger and Schumacher (2015), van der Ploeg (2014), van der Ploeg and de Zeeuw (2018), and Zemel (2015) view climate change as affecting the probability of a particular catastrophic event occurring, for example by modeling shock occurrence as a Poisson process whose arrival rate increases with pollution. However, there is less guidance on how to address the increased prevalence and intensity of "everyday" extreme weather events, which is a matter not only of uncertain timing but also uncertain impacts. This paper fills this gap by modeling the entire distribution of climate shocks as shifting continuously with the pollution stock, thus capturing the gradual degradation of the climate risk profile rather than an increasing probability of a sudden regime shift. To our knowledge, this second mechanism that is now well explored in the Earth sciences has received much less attention in economics. However, some recent work is being done to estimate the effect of climate change on the distribution of weather shocks and their economic impacts, such as Bakkensen and Barrage (2026). Therefore, this paper further bridges the micro-macro gap by using the insights from the emerging empirical literature on the effects of climate change on the distribution of weather shocks as well as the implications on micro-level damages to inform the macro-level implications on optimal policy.

The remainder of the paper is organized as follows. Section 2 introduces the three-period model, progressing from the familiar standard externalities case to endogenous stochastic externalities and then ending with a normal distribution and a Pareto example. Section 3 generalizes the analysis to an infinite-horizon recursive model and Section 4 applies it to a modified DICE benchmark model. Finally, Section 5 concludes.

2 Three-period model

2.1 Standard externalities

Let's start with the simplest case, where there are only three periods and output Y contributes to a pollution stock S which reduces a fraction $D(S) \in [0, 1)$ of next period's total output. Suppose damages are increasing and convex in the pollution stock. The planner solves:

$$\begin{aligned} \max_{\{c_0, c_1, c_2, k_1, k_2, S_1, S_2\}} \quad & u(c_0) + \beta u(c_1) + \beta^2 u(c_2) \\ \text{s.t.} \quad & \\ & c_0 + k_1 - (1 - \delta)k_0 = (1 - D(S_0))Y(k_0) \\ & c_1 + k_2 - (1 - \delta)k_1 = (1 - D(S_1))Y(k_1) \\ & c_2 - (1 - \delta)k_2 = (1 - D(S_2))Y(k_2) \\ & S_1 = (1 - \phi)S_0 + Y(k_0) \\ & S_2 = (1 - \phi)S_1 + Y(k_1) \end{aligned}$$

This problem has the following modified Euler condition between periods 0 and 1.

$$u'(c_0) = \beta u'(c_1) \left(1 - \delta + \left[1 - D(S_1) - \underbrace{\beta \frac{u'(c_2)}{u'(c_1)} D_S(S_2) Y(k_2)}_{\text{"Standard" externality}} \right] Y'(k_1) \right)$$

Since this is a standard problem, we do not solve the household's problem in detail. However, remember that the household's Euler condition does not contain the "standard" externality above and the planner would therefore like to set a carbon tax $\tau = \beta \frac{u'(c_2)}{u'(c_1)} D_S(S_2) Y(k_2)$ on output to internalize the externality. This is the standard Pigouvian tax, equal to the discounted marginal damages of pollution. Since second-period output is entirely determined by first-period investment, only the Euler condition between periods 0 and 1 is affected.

We may also consider the case where damages are a function of the pollution stock as well as a random shock x following an "exogenous" distribution, $D(S, x)$. Suppose damages are increasing and convex in the pollution stock in climate shocks. In that case, we get the following Euler condition, implying an optimal carbon tax $\tau = \beta \mathbb{E} \left[\frac{u'(c_2)}{u'(c_1)} D_S(S_2) Y(k_2) \right]$.

$$u'(c_0) = \beta \mathbb{E}_{x_1} \left[u'(c_1) \left(1 - \delta + \left[1 - D(S_1, x_1) - \underbrace{\beta \mathbb{E}_{x_2} \left[\frac{u'(c_2)}{u'(c_1)} D_S(S_2) Y(k_2) \right]}_{\text{"Standard" externality}} \right] Y'(k_1) \right) \right]$$

2.2 Stochastic externalities

Now consider the same model where future damages are stochastic, but the parameters of their distribution shifts with the pollution stock. We make the following hypotheses with regards to the stochastic process: (i) the shocks follow a statistical model of the parametric family which (ii) has fixed support, (iii) a pdf and cdf continuous and differentiable in parameter θ , (iv) a parameter θ continuous and differentiable in c_0 , and (v) a strictly increasing cdf. We represent the model's dependence on current consumption by treating the parameter $\theta(c_0)$ as a function of past consumption. The social planner solves:

$$\begin{aligned} \max_{\{c_0, c_1, c_2, k_1, k_2, S_1, S_2\}} \quad & u(c_0) + \beta \int u(c_1(x_1)) f(x_1; \theta(S_1)) dx_1 \\ & + \beta^2 \int \int u(c_2(x_2)) f(x_1; \theta(S_1)) f(x_2; \theta(S_2)) dx_1 dx_2 \\ \text{s.t.} \quad & \end{aligned} \tag{1}$$

$$\begin{aligned} c_0 + k_1 - (1 - \delta)k_0 &= (1 - D(S_0))Y(k_0) \\ c_1(x_1) + k_2(x_1) - (1 - \delta)k_1 &= (1 - D(S_1, x_1))Y(k_1) \quad \forall x_1 \\ c_2(x_2) - (1 - \delta)k_2(x_1) &= (1 - D(S_2, x_2))Y(k_2(x_1)) \quad \forall x_1, x_2 \\ S_1 &= (1 - \phi)S_0 + Y(k_0) \\ S_2 &= (1 - \phi)S_1 + Y(k_1) \end{aligned}$$

We write the expectations in integral form to make the dependence of the distribution on the pollution stock explicit which emerges in the PDF $f(x; \theta(S))$. Meanwhile, the consumer solves the following problem, where we denote by θ_1 and θ_2 the parameters of the distributions for x_1 and x_2 respectively.

$$\begin{aligned} \max_{\{c_0, c_1, c_2, k_1, k_2\}} \quad & u(c_0) + \beta \int u(c_1(x_1)) f(x_1; \theta_1) dx_1 + \beta^2 \int \int u(c_2(x_2)) f(x_1; \theta_1) f(x_2; \theta_2) dx_1 dx_2 \\ \text{s.t.} \quad & \end{aligned}$$

$$\begin{aligned} c_0 + k_1 - (1 - \delta)k_0 &= (1 + r_0)k_0 \\ c_1(x_1) + k_2(x_1) - (1 - \delta)k_1 &= r_1(x_1)k_1 \quad \forall x_1 \\ c_2(x_2) - (1 - \delta)k_2(x_1) &= r_2(x_2)k_2(x_1) \quad \forall x_1, x_2 \end{aligned}$$

The consumer takes the PDF as exogenous, hence us not treating θ as functions of pollution. This leads to the following modified expressions for the optimal corrective tax.

Proposition 2.2.1 (Three-period carbon tax with endogenous stochastic externalities).
The planner can decentralize the solution to problem (1) with the following carbon tax:

$$\tau = \underbrace{\beta \mathbb{E}_{x_1, x_2} \left[\frac{u'(c_2)}{u'(c_1)} D_S(S_2, x_2) Y(k_2) \right]}_{\text{"Standard" externality}} - \underbrace{\beta \text{Cov} \left(\frac{u(c_2)}{u'(c_1)}, \frac{\partial \ln(f(x_2; \theta(S_2)))}{\partial S_2} \right)}_{\text{"Stochastic" externality}}$$

This tax can alternatively be expressed as follows:

$$\tau = \underbrace{\beta \mathbb{E}_{x_1, x_2} \left[\frac{u'(c_2)}{u'(c_1)} D_S(S_2, x_2) Y(k_2) \right]}_{\text{"Standard" externality}} + \underbrace{\beta \mathbb{E}_{x_1, x_2} \left[\frac{u'(c_2)}{u'(c_1)} D_x(S_2, x_2) Y(k_2) \frac{\partial Q(p(x_2); \theta(S_2))}{\partial S_2} \right]}_{\text{"Stochastic" externality}}$$

Proof. See Appendix A for full proof. □

Looking at the formulas, we notice that our set up nests the standard externality. Indeed, we get the deterministic Pigouvian tax when the shocks follow a degenerate distribution and the Pigouvian tax in expectations when the shocks follow an exogenous distribution. With stochastic externalities, we get equation (7) with an additional term which implies a higher corrective tax when it is negative.

The first equation in Proposition 2.2.1 has a positive second term when utility and the score move in the same direction on average. We know that utility is monotone decreasing in shocks since higher shocks mean lower output and therefore lower consumption. Meanwhile, the score can be positive or negative depending on whether the probability of a certain shock realization increases or decreases after the probability shift.

The second equation is equivalent, but has a different interpretation. The first term is the stochastic discount factor, the second term is the marginal damages of shocks on output, and the third term is the change in the quantile function caused by the probability shift. Intuitively, if we for example think of the 3rd quartile, this says whether the position of this quartile goes up (75% of shocks are now higher) or down (75% of shocks are now lower).

To build some intuition on how the extra term works, we present two illustrative examples. Nevertheless, the framework we present is generalizable to any distribution as long as the distribution satisfies the assumptions mentioned at the start of the section.

Normal shocks. Let $x \sim N(\mu(S), \sigma(S))$ such that $\mu'(S) > 0$ (higher pollution increases average shocks) and $\sigma'(S) > 0$ (higher pollution increases shock variability).

Proposition 2.2.2 (Normal distribution decomposition).

Under normal shocks, the stochastic externality correction term decomposes into 3 channels:

$$\begin{aligned} -\mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} \frac{\partial \ln(f(\cdot))}{\partial S_2} \right] &= \mu'(S_2) \mathbb{E} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial D(\cdot)}{\partial x_2} Y(k_2) \right] \\ &\quad - \sigma'(S_2) \sigma(S_2) \mathbb{E} \left[\frac{u''(c_2)}{u'(c_1)} \left(\frac{\partial D(\cdot)}{\partial x_2} Y(k_2) \right)^2 \right] \\ &\quad + \sigma'(S_2) \sigma(S_2) \mathbb{E} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial^2 D(\cdot)}{\partial x_2^2} Y(k_2) \right] \end{aligned}$$

Proof. See Appendix B for full proof. □

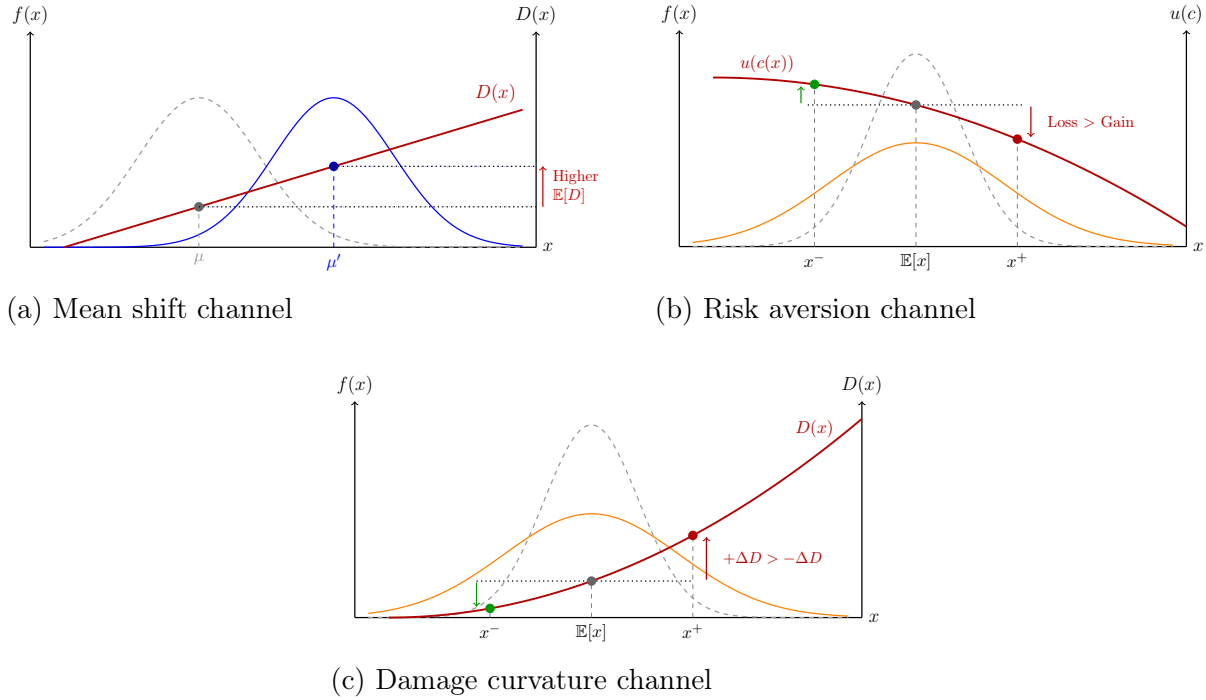


Figure 1: Stochastic externality channels for normally distributed shocks.

Figure 1 gives an intuition for the three channels. The mean shift channel in panel (a) leads to a higher carbon tax if damages are increasing since climate shocks are now higher on average. The risk aversion channel in panel (b) leads to a higher carbon tax if utility is concave because, although a variance increase makes both lower and higher shocks more likely, this causes a higher utility decline in the right-tails than utility gained in the left-tails.

If utility is linear such that the agent is risk-neutral, the risk aversion channel vanishes because both extremes become equally likely, yet the agent is indifferent between them. The damage curvature channel in panel (c) depends on the curvature of the damage function with respect to climate shocks. If damages are convex, meaning damages grow faster as shocks grow larger, this channel raises the tax because expanding the tails increases damages on the right tail more than it decreases damages on the left tail.

Pareto shocks. Let $x \sim \text{Pareto}(\alpha(S), x_m)$ with $\alpha'(S) < 0$ (higher pollution reduces the shape parameter, making the tail heavier) and fixed support $x_m > 0$. Moreover, suppose $\alpha > 1$ such that a finite first moment exists.

Proposition 2.2.3 (Pareto distribution decomposition).

Under Pareto shocks, the stochastic externality correction term can be rewritten as follows:

$$-\mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} \frac{\partial \ln(f(\cdot))}{\partial S_2} \right] = \frac{\alpha'(S_2)}{\alpha(S_2)} \mathbb{E} \left[\frac{u'(c_2)}{u'(c_1)} \frac{D(\cdot)}{\partial x_2} Y(k_2) x_2 \ln \left(\frac{x_2}{x_m} \right) \right]$$

Proof. The Pareto PDF is $f(x; \alpha) = \alpha x_m^\alpha x^{-(\alpha+1)}$ and CDF is $F(x; \alpha) = 1 - \left(\frac{x_m}{x}\right)^\alpha$ for $x \geq x_m$. We directly apply the result in 2.2.1 on the CDF form of the stochastic externality term. $\square$

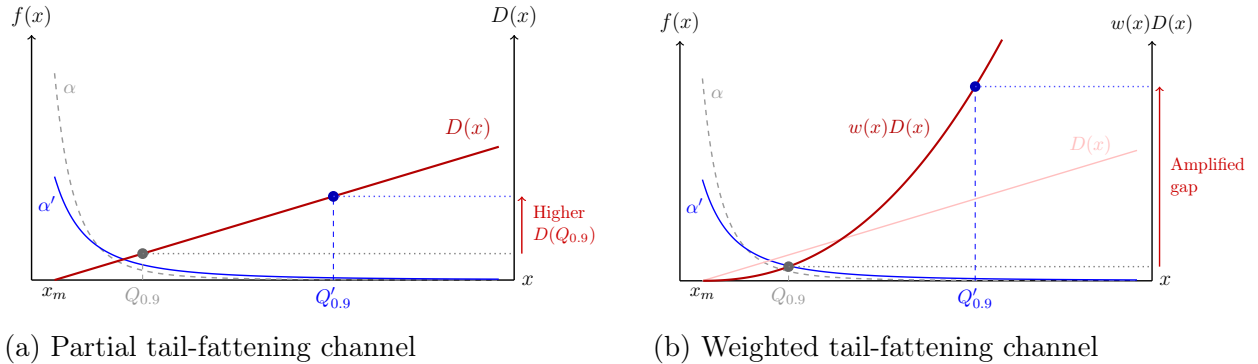


Figure 2: Pareto tail-fattening channel.

Endogenous Pareto shocks involve a single tail-fattening channel, as illustrated in figure 2. When the shape parameter α decreases and the right tail becomes heavier, each quantile of shocks shifts right as shown in panel (a). However, as shown in panel (b), right-tail damages are also weighted by a quantile shift term $w(x) \equiv x \ln(x/x_m)$ which grows exponentially, putting an increasing weight on rare catastrophic events. Depending on functional forms, the above expectation might diverge and imply an unbounded stochastic externality term, echoing the "Dismal Theorem" in Weitzman (2009). Indeed, the expectation is more "fragile" than the standard externality term. Since this paper is not focused on the Dismal Theorem, we will limit ourselves to a quick detour. Let us focus on $x \ln(x/x_m)$ and ignore the other terms in the expectation. If $\alpha > 1$, we have $\mathbb{E}[x] = \frac{\alpha x_m}{\alpha - 1}$. Now let us turn to the expectation of $x \ln(x/x_m)$ and note that $y \equiv \ln(x/x_m)$ follows an exponential distribution with rate α .

$$\mathbb{E} \left[x \ln \left(\frac{x}{x_m} \right) \right] = x_m \mathbb{E}[ye^y]$$

$\mathbb{E}[ye^y]$ is the derivative of the exponential distribution's moment generating function with respect to the moment parameter. Indeed,

$$M_y = \mathbb{E}[e^{sy}] \implies \frac{\partial M_y}{\partial s} = \mathbb{E}[ye^{sy}]$$

Since we can also write $M_y = \alpha/(\alpha - s)$,

$$M_y = \frac{\alpha}{\alpha - s} \implies \mathbb{E}[ye^y] = \left. \frac{\partial M_y}{\partial s} \right|_{s=1} = \frac{\alpha}{(\alpha - 1)^2}$$

Hence,

$$\mathbb{E} \left[x \ln \left(\frac{x}{x_m} \right) \right] = \frac{\alpha x_m}{(\alpha - 1)^2} = \frac{\mathbb{E}[x]}{\alpha - 1}$$

As $\alpha \rightarrow 1$ and the tails become heavier, $\mathbb{E}[x \ln(x/x_m)]$ diverges faster than $\mathbb{E}[x]$ such that the stochastic externality term may become unbounded before the standard externality term. Hence, even if we find that the Dismal Theorem does not hold with standard externalities¹, it may still hold with stochastic externalities. We leave further analysis of the implications of stochastic externalities on the Dismal Theorem for future work. However, the main takeaway is that the assumption on the distribution of shocks affects not only the channels by which stochastic externalities operate but also their importance relative to standard externalities, having the potential to dominate them. This is an important concern because the distribution of different climate events is hypothesized to be fat-tailed or heavy-tailed (generalized Gamma, generalized Pareto, etc.). Whether the aggregation of all types of climate shocks into a single stochastic damage function gives a fat-tailed or heavy-tailed distribution is an open question requiring further research.

In the next section, we generalize the simple three-period framework to an arbitrary number of periods. Before we move on, we want to stress that our general results are not a consequence of the specific way we have modeled the externality. We chose the standard neoclassical model with multiplicative damages on output to better build intuition, but we could have very well had modeled externalities as emerging from polluting consumption instead of polluting output. Though the specific form of the externality would be different, a stochastic externality would still arise with the same channels at play².

¹See for example Nordhaus (2009) or Anthoff and Tol (2022) which present the Dismal Theorem as a real concern but show its applicability depends on the hypotheses being made and the modeling context.

²For example, polluting consumption would affect the Euler condition through marginal utility rather than marginal product of output. However, we would still add a standard externality term and a stochastic externality term, with its channels depending on the assumption on the distribution of shocks.

3 Multiple period model

Now suppose an infinite horizon problem and suppose a pollution stock accumulates, with the past stock depreciating linearly at rate ϕ . In integral form, the sequential problem can be written in the following way, where the pdf of shocks depends on the pollution stock. Let $x^t \equiv (x_0, x_1, \dots, x_t)$ be the history of shocks up to t and $\beta \in (0, 1)$ be the discount factor.

$$\begin{aligned} \max_{\{c_t(x^t), k_{t+1}(x^t), S_{t+1}(x^t)\}_{t=0}^{\infty}} & \left\{ \sum_{t=0}^{\infty} \beta^t \int \cdots \int u(c_t(x^t)) f(x_0; \theta(S_0)) \cdots f(x_t; \theta(S_t)) dx_0 \cdots dx_t \right\} \quad (2) \\ \text{s.t.} & \end{aligned}$$

$$\begin{aligned} c_t(x^t) + k_{t+1}(x^t) &= (1 - D(S_t, x_t))Y(k_t(x^{t-1})) + (1 - \delta)k_t(x^{t-1}) & \forall t \\ S_{t+1}(x^t) &= (1 - \phi)S_t(x^{t-1}) + Y(k_t(x^{t-1})) & \forall t \\ x_{t+1} &\sim f(x_{t+1}; \theta(S_{t+1}(x^t))) & \forall t \end{aligned}$$

As written, this sequential problem is not very tractable. Hence, it can be useful to rewrite it recursively to make it more akin to the two-period problem. Although standard recursive methods as often used in economics assume exogenous shocks following a Markov process (see for example Stokey, Lucas, and Prescott (1989)), recursivity can still be established here. Indeed, our shocks essentially follow a *Markov decision process* (MDP) where the transition probability depends on the current state and the current action. With minor adjustments, the main results of SLP still hold³. First, let us make the following standard assumptions.

Assumption 3.0.1.

Let Γ be the correspondence for the constraints of this problem mapping states $s \in \mathbb{S}$ (i.e. capital and pollution stock) to actions $a \in \mathbb{A}$ (i.e. output). Suppose Γ has a measurable selection, meaning there is a measurable function $h : \mathbb{S} \rightarrow \mathbb{A}$ such that $h(s) \in \Gamma(s) \quad \forall s \in \mathbb{S}$.

Assumption 3.0.2.

Let utility $u : \mathbb{S} \times \mathbb{A} \rightarrow \mathbb{R}$ and the stochastic kernel $P : (\mathbb{S} \times \mathbb{A}) \times \mathcal{S} \rightarrow [0, 1]$, where $\mathcal{S}$ is the σ -algebra over the state space $\mathcal{S}$. Let a plan $\pi \in \Pi$ be a sequence of actions $(\pi_0, \pi_1, \dots)$ such that $\pi_0 \in \mathbb{A}$ and $\pi_t : \mathbb{S}^t \rightarrow \mathbb{A}$ are measurable for all t . For each $s_0 \in \mathbb{S}$ and each $\pi \in \Pi(s_0)$, $u(s_t, \pi_t(s^t))$ is $\mu^t(s_0, ds^t; \pi)$ -integrable and the following limit exists.

$$u(s_0, \pi_0) + \lim_{T \rightarrow \infty} \sum_{t=1}^T \beta^t \int u(s_t, \pi_t(s^t)) \mu^t(s_0, ds^t; \pi)$$

³See Miao (2013) for a simple introduction to MDP dynamic programming or, for more formal treatment, Bertsekas (1995), Hernández-Lerma and Lasserre (1996), etc.

Assumption 3.0.3.

Let there be a collection of non-negative measurable functions $L_t = \mathbb{S} \rightarrow \mathbb{R}_+ \forall t$ such that

$$u(s_0, \pi_0) \leq L_0(s_0) , \quad u(s_t, \pi_t(s^t)) \leq L_t(s_0) , \quad \sum_{t=0}^{\infty} \beta^t L_t(s_0) < \infty \quad \forall \pi \in \Pi(s_0), s_0 \in \mathbb{S}$$

Assumption 3.0.1 ensures a feasible policy exists for any state, Assumption 3.0.2 ensures that the discounted sum of utilities is well-defined, and Assumption 3.0.3 is a technical condition allowing for the use of the dominated convergence theorem in the following lemma.

Lemma 3.0.1 (Recursive formulation).

We can rewrite the sequential problem (2) as the following recursive problem.

$$\begin{aligned} V(k, S, x) = \max_{c, k', S'} & \left\{ u(c) + \beta \int V(k', S', x') f(x'; \theta(S')) dx' \right\} \\ & s.t. \\ c + k' &= (1 - D(S, x))Y(k) + (1 - \delta)k \\ S' &= (1 - \phi)S + Y(k) \\ x' &\sim f(x'; \theta(S')) \end{aligned} \tag{3}$$

Proof. We follow Miao (2013) for this proof, but other textbooks may be used as reference. It suffices to show that our setting is a standard MDP dynamic programming problem. Here, the state is the triple (k, S, x) and the action is (k', S') . Note that we can rewrite the constraints such that c is an implicit function of (k', S') . By Assumptions 3.0.1 and 3.0.2, we can define the following discounted expected utility for any given state s_0 and policy π .

$$J(s_0, \pi) = u(s_0, \pi_0) + \lim_{T \rightarrow \infty} \left\{ \sum_{t=1}^T \beta^t \int u(s_t, \pi_t(s^t)) \mu^t(s_0, ds^t; \pi) \right\}$$

Moreover, for any state, we can define the following value function.

$$V(s) = \sup_{\pi \in \Pi(s)} \{J(s, \pi)\}$$

With our particular notation, $s_0 = (k_0, S_0, x_0)$, $s_t = (k_t, S_t, x_t) \forall t$, and $s = (k, S, x)$. Here, $\pi \in \Pi(s)$ means that the policy π is feasible given the initial state s , where feasibility is established by the constraint set.

Suppose that $V : \mathbb{S} \rightarrow \mathbb{R}$ is measurable and, given any $s \in \mathbb{S}$ and any $\epsilon > 0$,

$$\exists \pi^s \in \Pi(s) : J(s, \pi) + \epsilon \geq V(s)$$

such that π_0^s and $\pi_t^s(s_1^t)$ are measurable for all t . By Theorem 7.1.1 of Miao (2013), Assumptions 3.0.1-3.0.3 together with the above assumption means V satisfies:

$$V(s) = \sup_{a \in \Gamma(s)} \left\{ u(s, a) + \beta \int V(s') P(s, a; ds') \right\}$$

In our notation,

$$V(k, S, x) = \max_{(k', S') \in \Gamma(k, S, x)} \left\{ u((k, S, x)) + \beta \int V(k', S', x') f(x'; S') dx' \right\}$$

Alternatively, noting that c is an implicit function of (k, S, x) ,

$$\begin{aligned} V(k, S, x) = \max_{c, k', S'} & \left\{ u(c) + \beta \int V(k', S', x') f(x'; \theta(S')) dx' \right\} \\ & \text{s.t.} \\ c + k' &= (1 - D(S, x))Y(k) + (1 - \delta)k \\ S' &= (1 - \phi)S + Y(k) \\ x' &\sim f(x'; \theta(S')) \end{aligned}$$

By Theorem 7.1.2 of Miao (2013), this equation is also equivalent to $V(s) = \sup_{\pi \in \Pi(s)} \{J(s, \pi)\}$, which means that the recursive problem is equivalent to the sequential problem. $\square$

Hence, although the shocks appear i.i.d., the dependence on last period's emissions creates persistence in the stochastic process. Since we assumed that consumption only affects the distribution of shocks through the pollution stock, the state variable S is sufficient to capture the effect of past emissions on future shocks. This gives the stochastic process a Markov structure, which allows us to write the problem in a recursive form. Using this result, we can derive the Euler condition for the multi-period model, generalizing the results from the two-period model to an arbitrary horizon.

Proposition 3.0.1 (Carbon tax with endogenous stochastic externalities).

The planner can decentralize the solution to problem (3) with the following carbon tax:

$$\tau_t = \underbrace{\beta \mathbb{E} \left[\sum_{j=0}^{\infty} \beta^j (1 - \phi)^j \frac{u'(c_{t+j+2})}{u'(c_{t+1})} D_{S,t+j+2} Y_{t+j+2} \right]}_{\text{"Standard" externality}} - \underbrace{\beta \sum_{j=0}^{\infty} \beta^j (1 - \phi)^j \text{Cov} \left(\frac{V_{t+j+2}}{u'(c_{t+1})}, \frac{\partial \ln(f_{t+j+2})}{\partial S_{t+j+2}} \right)}_{\text{"Stochastic" externality}}$$

This tax can alternatively be expressed as follows:

$$\tau_t = \underbrace{\beta \mathbb{E} \left[\sum_{j=0}^{\infty} \beta^j (1 - \phi)^j \frac{u'(c_{t+j+2})}{u'(c_{t+1})} D_{S,t+j+2} Y_{t+j+2} \right]}_{\text{"Standard" externality}} + \underbrace{\beta \mathbb{E} \left[\sum_{j=0}^{\infty} \beta^j (1 - \phi)^j \frac{u'(c_{t+j+2})}{u'(c_{t+1})} D_{x,t+j+2} Y_{t+j+2} \frac{\partial Q(p(x_{t+j+2}); \theta(S_{t+j+2}))}{\partial S_{t+j+2}} \right]}_{\text{"Stochastic" externality}}$$

Proof. See Appendix C for full proof. □

The above formula generalizes the three-period case to an arbitrary horizon. Because the polluting stock depreciates at rate ϕ , the effect of current emissions on future shocks is discounted by a factor $(1 - \phi)^j$ after j periods. That is, only $(1 - \phi)^j$ of emissions at time t will remain and harm the environment at time $t + j$. The first term is the standard Pigouvian term equal to the discounted expected marginal damages of pollution, while the second term is the "risk endogeneity" term capturing the effect of pollution on the distribution of shocks.

4 DICE-SE model

We implement the theoretical framework by modifying a single equation in DICE: the damage function. All other components—carbon cycle, climate dynamics, production, and the objective—follow the DSICE model of Cai and Judd (2023) unchanged, where DSICE itself extends the canonical DICE model of Nordhaus by incorporating stochastic shocks and solving the resulting dynamic optimization problem using the Simulated Certainty-Equivalent (SCEQ) algorithm. Note that the stochastic element in DSICE is a multiplicative TFP shock on gross output; we replace it with a damage shock x_t that enters only the damage denominator, shifting the source of uncertainty from productivity to climate damages. We introduce this shock first with an exogenous distribution as a baseline, and then with an endogenous distribution where the shock parameters depend on the climate state. We call the resulting model DICE-SE (stochastic externality).

4.1 Exogenous damage shocks

The DICE-SE model retains the six standard state variables: the capital stock K_t , three carbon reservoirs ($M_{AT,t}$, $M_{U,t}$, $M_{L,t}$), and two temperature variables ($T_{AT,t}$, $T_{OC,t}$). The carbon cycle and climate dynamics follow the standard DICE specification unchanged and are reproduced in Appendix D for reference. The capital stock K_t corresponds to the theoretical k , while the pollution stock S is captured by the full carbon-climate system; the theoretical damage function $D(S, x)$ becomes a function of T_{AT} and the stochastic shock x .

In the standard DICE model, climate damages reduce net output through a quadratic function of temperature. We modify this by adding a stochastic damage term. Gross output is $\tilde{Y}_t = A_t L_t^{1-\gamma} K_t^\gamma$ where A_t is total factor productivity, L_t is population, and $\gamma = 0.3$ is the capital share. Net output after abatement costs and climate damage is:

$$Y_t = \frac{(1 - \Lambda_t) \tilde{Y}_t}{1 + a_1 T_{AT,t} + a_2 T_{AT,t}^2 + \psi x_t}$$

where Λ_t is the abatement cost fraction, $a_1 = 0$ and $a_2 = 0.00284$ are the standard DICE damage coefficients, and $\psi = 0.1$ scales the stochastic damage shock x_t . The DICE coefficients are *unchanged*; the only addition is the term ψx_t . Setting $\phi = 0$ exactly recovers the standard deterministic DICE damage function. This maps to the theoretical damage function as $1 - D(S, x) = 1/(1 + a_1 T_{AT} + a_2 T_{AT}^2 + \psi x)$, so that higher shock values increase the damage fraction, reducing net output. The capital accumulation equation closes the model:

$$K_{t+1} = (1 - \delta_K) K_t + Y_t - C_t$$

where δ is the depreciation rate and C_t is aggregate consumption.

While the theoretical model assumes i.i.d. shocks for analytical clarity, the AR(1) strictly nests that case ($\rho = 0$) and better captures the empirical persistence of climate-related damage events. The stochastic externality formula extends unchanged: the relevant object is the conditional score $\partial \ln f(x_t | x_{t-1}; S) / \partial S$, which for the AR(1) is given by equation (6) in Section 4.4.

The damage shock follows an AR(1) process:

$$x_{t+1} = \rho x_t + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_\varepsilon^2)$$

with persistence $\rho = 0.8$ and innovation standard deviation $\sigma_\varepsilon = 0.3$, so that the unconditional standard deviation is $\sigma_x = \sigma_\varepsilon / \sqrt{1 - \rho^2} = 0.5$. We discretize this process using the method of **tauchen_finite_1986**<empty citation> with $K = 20$ grid points spanning $\pm 3\sigma_x$; the transition probability formula is given in Appendix D.

The choice of $\rho = 0.8$ matches the persistence of the TFP shock in the original DSICE model of Cai and Judd (2023), which uses a 3-state Markov chain with values $\{0.9, 1.0, 1.1\}$ and staying probability 0.8. Our 20-state discretization preserves this persistence while providing a richer grid that can accommodate the endogenous distributional shifts introduced in Section 4.4. The innovation standard deviation $\sigma_\varepsilon = 0.3$ yields an unconditional standard deviation $\sigma_x = 0.5$, producing a grid that spans well beyond the original ± 0.1 range and captures the full distributional shape needed for computing the stochastic externality. The damage sensitivity parameter $\psi = 0.1$ is chosen so that a one-standard-deviation shock ($x = 0.5$) adds $\psi \cdot 0.5 = 0.05$ to the damage denominator, a meaningful but not dominant perturbation relative to the baseline quadratic damage $a_2 T_{AT}^2$.

4.2 SCEQ algorithm

We solve the model using the Simulated Certainty-Equivalent (SCEQ) algorithm of Cai and Judd (2023). The key idea is to replace future stochastic shocks with their conditional expectations, reducing the stochastic optimization to a sequence of deterministic nonlinear programs (NLPs). For $N = 167$ independent simulation paths, the algorithm proceeds at each period s :

1. **Fix current states.** For each path n , fix the realized state variables $(K_s^n, M_{AT,s}^n, \dots, x_s^n)$ at their current values.
2. **Compute certainty-equivalent shocks.** Given the current shock state $x_s = x_i$ on the Tauchen grid, compute the conditional expectation of future shocks:

$$\tilde{x}_t = \sum_{j=1}^K \pi_j^{(t)} x_j, \quad \text{where } \boldsymbol{\pi}^{(t)} = (P^\top)^{t-s} \mathbf{e}_i$$

and $\mathbf{e}_i$ is the i -th unit vector.

3. **Solve the deterministic NLP.** With states fixed and future shocks replaced by $\{\tilde{x}_t\}_{t>s}$, solve the deterministic optimization over the remaining horizon to obtain optimal decisions (μ_s^n, C_s^n) and next-period states.
4. **Sample next shock.** Draw the period- $(s+1)$ shock state from the transition matrix and advance to $s+1$. In the exogenous model, all N shock paths are pre-simulated from the fixed P before the SCEQ loop begins.⁴

⁴In the endogenous model, the draw must be made *after* each NLP solve, since $P(S_{\text{eff},t})$ depends on the realized climate trajectory and is only known post-solve. This sequential structure ensures that each path constitutes a valid draw from the equilibrium distribution, as discussed in the tax decomposition of Section 4.4.

4.3 Tax computation via impulse response

The optimal carbon tax at period s is computed via an impulse response function (IRF). A unit perturbation to atmospheric carbon M_{AT} at period $s + 1$ propagates through the carbon cycle (8) and climate dynamics (9), generating a temperature response $\{\delta T_{\text{AT},t}\}_{t>s}$. The standard Pigouvian tax is the discounted sum of marginal damages along this impulse response:

$$\tau_s^{\text{std}} = \sum_{t=s+1}^T \beta^{t-s} \frac{u'(c_t)}{u'(c_s)} \cdot \frac{(a_1 + 2a_2 T_{\text{AT},t}) \delta T_{\text{AT},t}}{1 + a_1 T_{\text{AT},t} + a_2 T_{\text{AT},t}^2 + \psi \tilde{x}_t} \cdot Y_t \quad (4)$$

where $u'(c) = c^{-\gamma}$ is the marginal utility of per-capita consumption and $\delta T_{\text{AT},t}$ is the impulse response of atmospheric temperature to the unit emission perturbation. The normalization by $u'(c_s)$ is absorbed into the stochastic discount factor $u'(c_t)/u'(c_s)$; in the endogenous extension it will appear explicitly in τ_s^{stoch} , where the continuation value is not pre-divided by marginal utility.

Since the transition matrix P is constant and independent of the climate state in the exogenous model, emissions have no effect on the distribution of future shocks. The socially optimal carbon tax therefore coincides with the standard Pigouvian component: $\tau_s = \tau_s^{\text{std}}$. This serves as the benchmark against which the stochastic externality premium in Section 4.4 is measured.

4.4 Endogenous damage shocks

We now extend the model so that the distribution of damage shocks depends on the climate state, implementing the theoretical framework of Sections 3 and 4. The key modification is that the parameters of the AR(1) innovation distribution become functions of an effective pollution state, creating a feedback loop between climate outcomes and the risk environment.

The use of realized-path continuation values rather than conditional expectations requires justification. The SCEQ algorithm embeds a "naïve social planner" who optimizes each period using certainty-equivalent shocks, treating the transition matrix as fixed and therefore ignoring that current emissions shift $P(S_{\text{eff}})$. The NLP tax τ^{NLP} is this planner's carbon price: it internalizes direct climate damages but misses the stochastic externality. By backward-recurring on the realized paths generated under the naïve planner's decisions and applying the score-weighted impulse response, we backtrack the welfare cost of the distributional channel he neglected, obtaining τ^{stoch} . The validity of this approach rests on the endogenous model's sequential structure: because $P(S_{\text{eff},t})$ depends on the endogenous climate trajectory, shock states cannot be pre-simulated and must instead be drawn on-the-fly from the true transition matrix after each NLP solve. Each path therefore constitutes a genuine draw from the equilibrium distribution, and averaging $\tau_s^{\text{stoch},n}$ across $N = 167$ paths recovers the required expectation by the law of large numbers. The corrected optimal tax $\tau_s = \tau_s^{\text{std}} + \tau_s^{\text{stoch}}$ is what a planner who fully internalizes both the direct damage channel and the distributional risk channel should levy.

4.4.1 Effective pollution state

We define the effective pollution state as a weighted combination of climate variables:

$$S_{\text{eff},t} = w_1 T_{\text{AT},t} + w_2 M_{\text{AT},t}$$

with $w_1 = 1$ and $w_2 = 0$, so that $S_{\text{eff},t} = T_{\text{AT},t}$ in the current calibration. This maps the theoretical pollution stock S to the atmospheric temperature anomaly, which is the primary driver of climate damages. The specification is flexible: nonzero w_2 would allow CO_2 concentration to directly affect the shock distribution.

Then, the AR(1) shock process now has state-dependent innovations:

$$x_{t+1} \mid x_t, S_{\text{eff},t} \sim N(\rho x_t + \mu(S_{\text{eff},t}), \sigma(S_{\text{eff},t})^2) \quad (5)$$

where the conditional mean and standard deviation of innovations depend on the effective pollution state:

$$\mu(S) = \mu_0 + \mu_1 S, \quad \sigma(S) = \sigma_0 \exp(\sigma_1 S)$$

with $\mu_0 = 0$, $\mu_1 = 0.1$, $\sigma_0 = 0.3$, and $\sigma_1 = 0.20$. This captures two distinct effects through which climate change affects the damage risk:

- *Mean-shift effect* ($\mu_1 > 0$): higher temperatures raise the expected damage shock, shifting the distribution toward more damaging states.
- *Volatility channel* ($\sigma_1 > 0$): higher temperatures increase the variance of damage shocks, fattening the tails and increasing the probability of extreme outcomes.

These effects bring forward the mean-shift, risk-aversion and the damage curvature channels, identified in the normal distribution decomposition of Proposition 2.2.2.

Figure 3 illustrates how the underlying normal distribution shifts across the simulation horizon. At each period t and path n , the Tauchen mechanism draws from the conditional distribution $N(\rho x_t^n + \mu(S_{\text{eff},t}^n), \sigma(S_{\text{eff},t}^n)^2)$. Since paths differ in their realized temperature and shock trajectories, each path implies a distinct normal at any given period. The curves in Figure 3 are the path-averaged mixture densities, obtained by evaluating the conditional pdf for each of the $N_{\text{sim}} = 167$ paths and averaging pointwise.

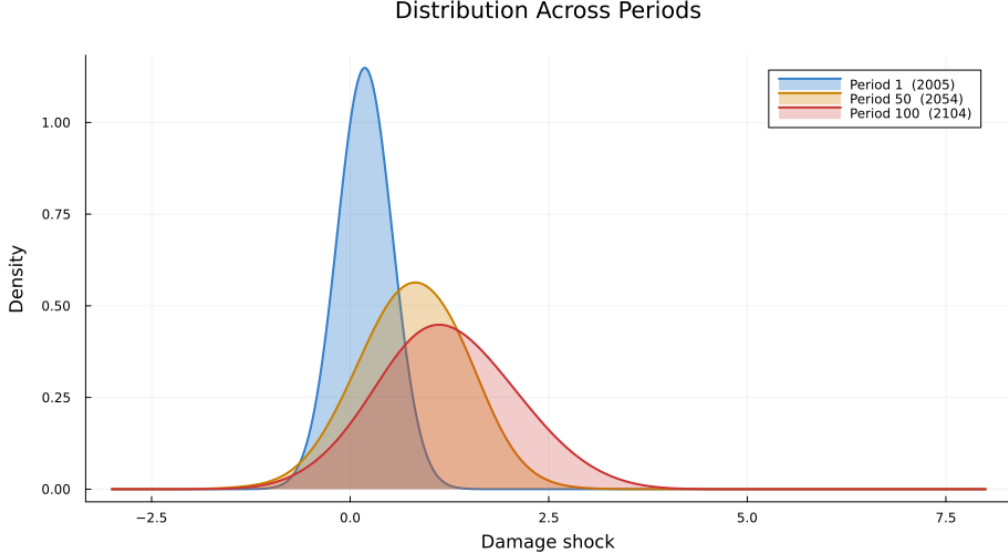


Figure 3: Each curve is the pointwise average of the 167 path-specific normals $N(\rho x_t^n + \mu(S_{\text{eff},t}^n), \sigma(S_{\text{eff},t}^n)^2)$ at the respective period.

The rightward shift reflects the mean-shift channel ($\mu_1 > 0$): as temperatures rise along simulated paths, $\mu(S_{\text{eff}})$ increases, pushing the distribution toward higher damage shocks. The widening of the distribution reflects the volatility channel ($\sigma_1 > 0$): higher S_{eff} raises $\sigma(S_{\text{eff}})$, fattening the tails and increasing the probability of extreme outcomes. As we've established, the changes in standard deviations will combine two effects, the risk aversion channel and the damage curvature channel. Together, these three channels correspond directly to the stochastic externality.

4.4.2 Tax decomposition

The optimal carbon tax decomposes into standard and stochastic components, as derived in Proposition 3.0.1:

$$\tau_s = \tau_s^{\text{std}} + \tau_s^{\text{stoch}}$$

The standard component τ_s^{std} is inherited unchanged from equation (4): the impulse response of temperatures to a unit emission perturbation is the same, and the direct damage channel

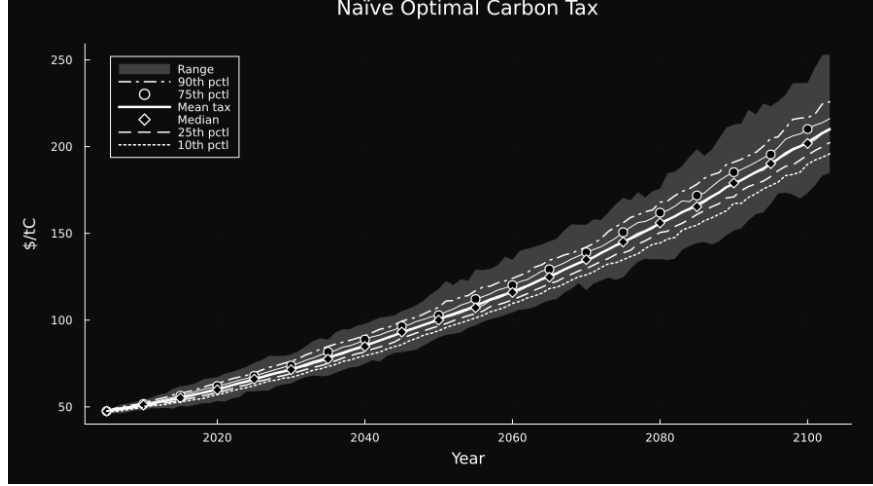


Figure 4: Naïve optimal carbon tax (τ^{std} , marginal abatement cost) across 167 simulation paths.

$(a_1 + 2a_2 T_{AT,t})$ operates identically. Figure 4 shows what the "Naïve Social Planner" would be establishing as the taxation.

What changes is the addition of a stochastic externality: emissions now also shift the distribution of future damage shocks through the state-dependent transition matrix $P(S_{\text{eff}})$. A unit emission perturbation at period s shifts $M_{AT,s+1}$, which propagates through the climate system and eventually perturbs the effective pollution state $S_{\text{eff},t}$ at future periods. This distributional shift is valued using the score function.

For the conditional normal distribution in (5), the score with respect to the effective pollution state is:

$$\frac{\partial \ln f(x_t | x_{t-1}; S_{\text{eff},t})}{\partial S_{\text{eff}}} = \frac{\mu_1}{\sigma(S_{\text{eff},t})} z_t + \sigma_1 (z_t^2 - 1) \quad (6)$$

where $z_t = (x_t - \rho x_{t-1} - \mu(S_{\text{eff},t}))/\sigma(S_{\text{eff},t})$ is the standardized innovation. The first term captures the mean-shift effect and the second term captures the volatility effect, mirroring the decomposition in Proposition 2.2.2.

The stochastic externality tax is then computed via impulse response:

$$\tau_s^{\text{stoch}} = -\frac{1}{u'(c_s)} \sum_{t=s+1}^T \beta^{t-s} \delta S_{\text{eff},t} \cdot V_t^{\text{cont}} \cdot \frac{\partial \ln f(x_t | x_{t-1}; S_{\text{eff},t})}{\partial S_{\text{eff}}}$$

where $\delta S_{\text{eff},t} = w_1 \delta T_{AT,t} + w_2 \delta M_{AT,t}$ is the impulse response of the effective pollution state to the unit emission perturbation, and $V_t^{\text{cont},n}$ is the realized continuation welfare along simulation path n , computed by backward recursion $V_t^{\text{cont},n} = u(c_t^n) + \beta V_{t+1}^{\text{cont},n}$ over the consumption path already produced by the SCEQ loop, requiring no additional NLP solves. This is the discrete-time, AR(1) analogue of the theoretical stochastic externality $-\beta \sum_j \beta^j (1-\phi)^j \text{Cov}(V/u'(c), \partial \ln f / \partial S)$ from Proposition 3.0.1, where the covariance is evaluated path-by-path and the depreciation factor $(1-\phi)^j$ is replaced by the explicit impulse response $\delta S_{\text{eff},t}$ through the carbon-climate system.

Figure 5 identifies across time the different cost of the stochastic externalities, which are larger in earlier periods due to the continuation value. In essence, a lack of consideration at the beginning comes at a higher cost across time.

Where, using our backward progation, we estimate a cost to the stochastic externality that

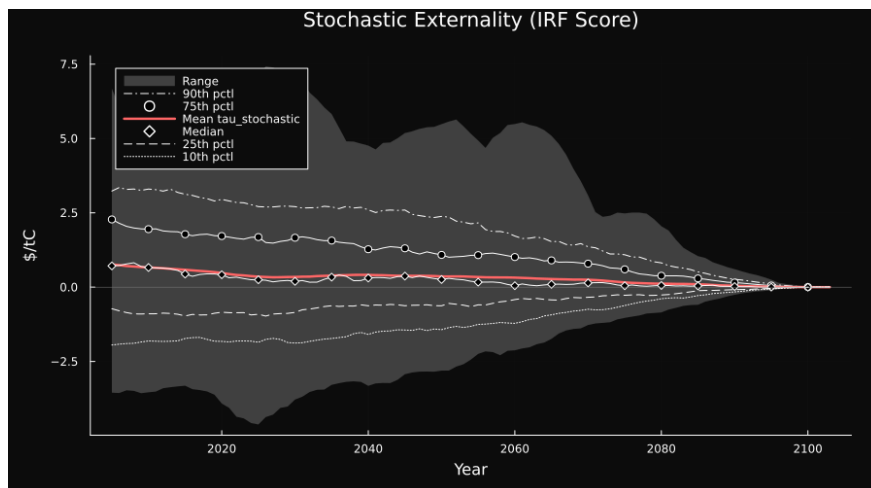


Figure 5: Stochastic externality component τ^{stoch} across simulation paths.

5 Conclusion

In this paper, we develop a framework for analyzing the optimal correction of stochastic externalities, defined as the effect of pollution on the distribution of future climate shocks. This reflects the "distributional shift" perspective of climate risk according to which higher CO₂ concentrations (or other climate variables) make extreme events not only more frequent but also more severe and more unpredictable is well-documented in the Earth sciences and is beginning to be empirically assessed in economics, yet its implications for the social cost of carbon and optimal climate policy have not been formally characterized. This paper fills this gap and offers avenues for further research.

Working first in a tractable three-period neoclassical model, we derive a modified optimal carbon tax that decomposes into two terms: the standard Pigouvian term and a novel stochastic externality term. The latter can be expressed as the covariance between the ratio of utilities and the score of the damage shock distribution, or equivalently as the discounted expectation of marginal damage shocks weighted by the change in the shock distribution's quantile function induced by the pollution-driven distributional shift. The framework nests the standard Pigouvian tax as a special case when the damage distribution is exogenous. Under normal shocks with pollution-dependent mean and variance, the stochastic externality decomposes into three economically interpretable channels: a mean-shift channel, a risk-aversion channel, and a damage-curvature channel. Under Pareto shocks, the correction collapses to a single tail-fattening channel whose magnitude can grow without bound as the tail becomes heavier, hinting that Weitzman's Dismal Theorem may be reinforced in the case of stochastic externalities. Indeed, even if the standard Pigouvian term remains finite, the stochastic externality term may not. The qualitative and quantitative results are sensitive to assumptions on the distribution of shocks and its parameters, highlighting the importance of rigorously calibrating the stochastic component of the model. We then generalize these results to an infinite-horizon recursive model.

To illustrate the quantitative importance of the stochastic externality, we implement these results in DICE-SE, a modified version of the DSICE model developed in Cai and Lontzek (2019) which is solved using a modified form of the SCEQ algorithm developed in Cai and Judd (2023). Although the full DSICE model also features TFP shocks and tipping points, we strip away these additional elements and focus solely on the baseline recursive DICE model it is based on. We then add a stochastic damage component to the initially deterministic damage function whose parameters shift with the climate stock (temperature or CO₂ concentration). In the present version of the paper, the distributional forms and parameters are only illustrative. However, further work will use a more realistic form of damage and use the climate science literature and the burgeoning empirical economic literature on the distribution shifting channel of climate risk. Nevertheless, when we compare the standard externalities case to the endogenous externalities case, the numerical exercise reveals that the theoretical results are consistent with the DICE-SE simulation. Once properly calibrated, the model will be able to give a more accurate estimate of the stochastic externality premium. Since some types of climate events are hypothesized to follow heavy or fat-tailed distributions, there is a potential for a large stochastic externality premium. Whether this is indeed the case depends on the assumption on the distribution of shocks.

This framework carries direct implications for climate policy. Standard integrated assessment models, which either treat climate damages as deterministic or merely focus on the timing of shocks may systematically underestimate the social cost of carbon. While the tipping points literature focuses on the possibility of large irreversible jumps in damages in the future, our distribution shifting mechanism instead focuses on the change in everyday climate shocks.

This work requires many improvements. Currently, the DICE-SE simulation is merely illustrative, using a normal distribution with arbitrary parameters rather than a carefully-chosen realistic shock distribution (e.g. GEV, etc.), as supported by the existing empirical literature. Future work will need to properly calibrate the DICE-SE model to provide a more accurate estimate of the stochastic externality premium.

There are also a number of important extensions for future work, such as extending the framework to heterogeneous agents to allow the study of the distributional consequences of taking into account stochastic externalities. Moreover, it could be very interesting to investigate the interaction between mitigation and adaptation in this framework, as adaptation may reduce the need to internalize the stochastic externality through mitigation. Likewise, internalizing stochastic externalities may reduce the need for adaptation, acting as global insurance. Following Schumacher (2019), one might even argue that mitigating stochastic externalities is superior to adapting to them, since the former is essentially a public good while the latter is a private good, only a few being able to benefit from adaptation. In other words, a stochastic externality framework combined with an insurance component may also help us evaluate the relative merits of *ex post* mitigation ("prevention is better than cure") and of *ex ante* adaptation in the context of climate change.

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A Proof of Proposition 2.2.1

First, we solve the household's problem, which has the following Lagrangian.

$$\begin{aligned}\mathcal{L} = & u(r_0 k_0 - k_1 + (1 - \delta)k_0) + \beta \int_{x_1} u(r_1(x_1)k_1 - k_2(x_1) + (1 - \delta)k_1) f(x_1; \theta_1) dx_1 \\ & + \beta^2 \int_{x_1} \int_{x_2} u(r_2(x_2)k_2(x_1) + (1 - \delta)k_2(x_1)) f(x_1; \theta_1) f(x_2; \theta_2) dx_1 dx_2\end{aligned}$$

We assume the household is atomistic and therefore does not take into account the effect of their actions on the environment. Hence, we treat the pdf as exogenous in this problem. This yields the following first-order conditions.

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial k_1} : u'(c_0) &= \beta \int_{x_1} u'(c_1)(1 - \delta + r_1) f(x_1; \theta_1) dx_1 \\ \frac{\partial \mathcal{L}}{\partial k_2} : u'(c_1) &= \beta \int_{x_2} u'(c_2)(1 - \delta + r_2) f(x_2; \theta_2) dx_2\end{aligned}$$

By the firm's optimality condition, we have $r_t = (1 - D(S_t, x_t))Y'(k_t)$. Hence,

$$\begin{aligned}u'(c_0) &= \beta \mathbb{E}_{x_1} [u'(c_1)(1 - \delta + (1 - D(S_1, x_1))Y'(k_1))] \\ u'(c_1) &= \beta \mathbb{E}_{x_2} [u'(c_2)(1 - \delta + (1 - D(S_2, x_2))Y'(k_2))]\end{aligned}$$

This is the standard Euler condition. The planner's problem yields the following FOCs.

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial k_1} : u'(c_0) &= \beta \int_{x_1} u'(c_1)(1 - \delta + (1 - D(S_1, x_1))Y'(k_1)) f(x_1; \theta) dx_1 + \mu_2 Y'(k_1) \\ \frac{\partial \mathcal{L}}{\partial k_2} : u'(c_1) &= \beta \int_{x_2} u'(c_2)(1 - \delta + (1 - D(S_2, x_2))Y'(k_2)) f(x_2; \theta) dx_2 \\ \frac{\partial \mathcal{L}}{\partial S_1} : & -\beta \int_{x_1} u'(c_1) D_S(S_1) Y(k_1) f(x_1; \theta) dx_1 \\ & + \beta \int_{x_1} u(c_1) \frac{\partial f(x_1; \theta)}{\partial \theta} \frac{\partial \theta(S_1)}{\partial S_1} dx_1 \\ & + \beta^2 \int_{x_1} \int_{x_2} u(c_2) \frac{\partial f(x_1; \theta)}{\partial \theta} \frac{\partial \theta(S_1)}{\partial S_1} f(x_2; \theta(S_2)) dx_1 dx_2 + (1 - \phi) \mu_2 = \mu_1 \\ \frac{\partial \mathcal{L}}{\partial S_2} : & -\beta^2 \int_{x_1} \int_{x_2} u'(c_2) D_S(S_2) Y(k_2) f(x_1; \theta) f(x_2; \theta) dx_1 dx_2 \\ & + \beta^2 \int_{x_1} \int_{x_2} u(c_2) \frac{\partial f(x_2; \theta)}{\partial \theta} \frac{\partial \theta(S_2)}{\partial S_2} f(x_1; \theta(S_1)) dx_1 dx_2 = \mu_2\end{aligned}$$

So that the terms in the brackets of the first-order condition with respect to c_0 are expressed as an expectation, we would like to rewrite the equation so that the second term is multiplied by the original pdf. To do so, we can use the property that $\frac{d \ln(f(x))}{dx} = \frac{f'(x)}{f(x)}$.

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial k_1} : u'(c_0) &= \beta \int_{x_1} u'(c_1)(1 - \delta + (1 - D(S_1, x_1))Y'(k_1))f(x_1; \theta)dx_1 + \mu_2 Y'(k_1) \\
\frac{\partial \mathcal{L}}{\partial k_2} : u'(c_1) &= \beta \int_{x_2} u'(c_2)(1 - \delta + (1 - D(S_2, x_2))Y'(k_2))f(x_2; \theta)dx_2 \\
\frac{\partial \mathcal{L}}{\partial S_1} : & -\beta \int_{x_1} u'(c_1)D_S(S_1)Y(k_1)f(x_1; \theta)dx_1 \\
& + \beta \int_{x_1} u(c_1) \frac{\partial \ln(f(x_1; \theta))}{\partial \theta} \frac{\partial \theta(S_1)}{\partial S_1} f(x_1; \theta(S_1))dx_1 \\
& + \beta^2 \int_{x_1} \int_{x_2} u(c_2) \frac{\partial \ln(f(x_1; \theta))}{\partial \theta} \frac{\partial \theta(S_1)}{\partial S_1} f(x_1; \theta(S_1))f(x_2; \theta(S_2))dx_1 dx_2 + (1 - \phi)\mu_2 = \mu_1 \\
\frac{\partial \mathcal{L}}{\partial S_2} : & -\beta^2 \int_{x_1} \int_{x_2} u'(c_2)D_S(S_2)Y(k_2)f(x_1; \theta)f(x_2; \theta)dx_1 dx_2 \\
& + \beta^2 \int_{x_1} \int_{x_2} u(c_2) \frac{\partial \ln(f(x_2; \theta))}{\partial \theta} \frac{\partial \theta(S_2)}{\partial S_2} f(x_1; \theta(S_1))f(x_2; \theta(S_2))dx_1 dx_2 = \mu_2
\end{aligned}$$

The last FOC can be written more compactly as follows.

$$\begin{aligned}
\mu_2 &= -\beta^2 \int_{x_1} \int_{x_2} u'(c_2)D_S(S_2)Y(k_2)f(x_1; \theta)f(x_2; \theta)dx_1 dx_2 \\
& + \beta^2 \int_{x_1} \int_{x_2} u(c_2) \frac{\partial \ln(f(x_2; \theta))}{\partial \theta} \frac{\partial \theta(S_2)}{\partial S_2} f(x_1; \theta)f(x_2; \theta)dx_1 dx_2
\end{aligned}$$

From here, we can get the following modified Euler condition between c_0 and c_1 .

$$u_{c,0} = \beta \mathbb{E}_{x_1} \left[u_{c,1} \left(1 - \delta + \left[1 - D_1 - \beta \mathbb{E}_{x_2} \left[\underbrace{\frac{u_{c,2}}{u_{c,1}} D_{S,2} Y_2}_{\text{"Standard" ext.}} - \underbrace{\frac{u(c_2)}{u_{c,1}} \frac{\partial \ln(f(\cdot))}{\partial S_2}}_{\text{"Stochastic" ext.}} \right] Y_{k,1} \right) \right] \quad (7)$$

If we compare this to the household's Euler condition, we see that there is a wedge on the right-hand side. In order to decentralize the planner's solution, we need to set a carbon tax

$$\tau = \beta \mathbb{E}_{x_1, x_2} \left[\underbrace{\frac{u'(c_2)}{u'(c_1)} D_S(S_2, x_2) Y(k_2)}_{\text{"Standard" externality}} - \underbrace{\frac{u(c_2)}{u'(c_1)} \frac{\partial \ln(f(x_2; \theta(S_2)))}{\partial S_2}}_{\text{"Stochastic" externality}} \right]$$

We can also write this in a covariance formulation⁵.

$$\tau = \underbrace{\beta \mathbb{E}_{x_1, x_2} \left[\frac{u'(c_2)}{u'(c_1)} D_S(S_2, x_2) Y(k_2) \right]}_{\text{"Standard" externality}} - \underbrace{\beta \text{Cov} \left(\frac{u(c_2)}{u'(c_1)}, \frac{\partial \ln(f(x_2; \theta(S_2)))}{\partial S_2} \right)}_{\text{"Stochastic" externality}}$$

We can also write the Euler condition in cdf terms rather than pdf terms by rewriting the first-order condition with respect to S_2 with integration by parts⁶.

$$\begin{aligned} \int_{\underline{x}}^{\bar{x}} u(c_2) \frac{\partial \theta(S_2)}{\partial S_2} \frac{\partial f(x_2; \theta(S_2))}{\partial \theta} dx_2 &= u(c_2) \frac{\partial \theta(S_2)}{\partial S_2} \frac{\partial F(x_2; \theta(S_2))}{\partial \theta} \Big|_{\underline{x}}^{\bar{x}} \\ &\quad - \int_{\underline{x}}^{\bar{x}} u'(c_2) \frac{\partial c_2}{\partial x_2} \frac{\partial \theta(S_2)}{\partial S_2} \frac{\partial F(x_2; \theta(S_2))}{\partial \theta} dx_2 \end{aligned}$$

By the Leibniz integral rule, the derivative of the cdf with respect to its parameter is zero at the boundaries. Therefore,

$$\int_{\underline{x}}^{\bar{x}} u(c_2) \frac{\partial \theta(S_2)}{\partial S_2} \frac{\partial f(x_2; \theta(S_2))}{\partial \theta} dx_2 = - \int_{\underline{x}}^{\bar{x}} u'(c_2) \frac{\partial c_2}{\partial x_2} \frac{\partial \theta(S_2)}{\partial S_2} \frac{\partial F(x_2; \theta(S_2))}{\partial \theta} dx_2$$

From the resource constraint, $\frac{\partial c_2}{\partial x_2} = -D_x(S_2, x_2)Y(k_2)$.

$$\int_{\underline{x}}^{\bar{x}} u(c_2) \frac{\partial \theta(S_2)}{\partial S_2} \frac{\partial f(x_2; \theta(S_2))}{\partial \theta} dx_2 = \int_{\underline{x}}^{\bar{x}} u'(c_2) D_x(S_2, x) Y(k_2) \frac{\partial \theta(S_2)}{\partial S_2} \frac{\partial F(x_2; \theta(S_2))}{\partial \theta} dx_2$$

⁵ $\frac{\partial \ln(f(\cdot))}{\partial \theta}$ is the score and $\mathbb{E}[S_\theta] = 0$. Hence, the second term in $\text{Cov}(x, y) = \mathbb{E}[xy] - \mathbb{E}[x] \mathbb{E}[y]$ cancels out.

⁶ Here, $u = u(c_2) \frac{\partial \theta(S_2)}{\partial S_2}$ and $dv = \frac{\partial f(\cdot)}{\partial \theta} dx$. Moreover, note that $\frac{\partial f(\cdot)}{\partial \theta} = \frac{\partial^2 F(\cdot)}{\partial x \partial \theta}$.

If we multiply and divide by $f(x_2; \theta(S_2)) \equiv \frac{\partial F(x_2; \theta(S_2))}{\partial x_2}$, we can rewrite equation (7) as:

$$u_{c,0} = \beta \mathbb{E}_{x_1} \left[u_{c,1} \left(1 - \delta + \left[1 - D_1 - \beta \mathbb{E}_{x_2} \left[\underbrace{\frac{u_{c,2}}{u_{c,1}} D_{S,2} Y_2}_{\text{"Standard" ext.}} - \underbrace{\frac{u_{c,2}}{u_{c,1}} D_x(S_2, x_2) Y(k_2) \frac{\frac{\partial F(x_2; \theta(S_2))}{\partial S_2}}{\frac{\partial F(x_2; \theta(S_2))}{\partial x_2}}}_{\text{"Stochastic" externality}} \right] \right] Y_{k,1} \right) \right]$$

There are three parts to this new term: the familiar stochastic discount factor, the marginal damages of stochastic shocks on output, and the change of the cdf over the pdf. This last part is hard to interpret, so we might want to rewrite it. The quantile function is defined as:

$$Q(p; \theta(S_2)) = F^{-1}(p; \theta(S_2))$$

From here, we can also write a cdf as follows.

$$F(Q(p; \theta(S_2)); \theta(S_2)) = p$$

If we differentiate both sides by S_2 and rearrange terms,

$$\frac{\frac{\partial F(Q(p; \theta(S_2)); \theta(S_2))}{\partial \theta} \frac{\partial \theta(S_2)}{\partial S_2}}{f(Q(p; \theta(S_2)); \theta(S_2))} = - \frac{\partial Q(p; \theta(S_2))}{\partial \theta} \frac{\partial \theta(S_2)}{\partial S_2}$$

If we set $x = Q(p; \theta(S_2))$ in the cdf,

$$\frac{\frac{\partial F(x; \theta(S_2))}{\partial \theta} \frac{\partial \theta(S_2)}{\partial S_2}}{f(x; \theta(S_2))} = - \frac{\partial Q(p; \theta(S_2))}{\partial \theta} \frac{\partial \theta(S_2)}{\partial S_2}$$

Hence, a different way to write equation (7) is:

$$u_{c,0} = \beta \mathbb{E}_{x_1} \left[u_{c,1} \left(1 - \delta + \left[1 - D_1 - \beta \mathbb{E}_{x_2} \left[\underbrace{\frac{u_{c,2}}{u_{c,1}} D_{S,2} Y_2}_{\text{"Standard" ext.}} - \underbrace{\frac{u_{c,2}}{u_{c,1}} D_x(S_2, x_2) Y(k_2) \frac{\partial Q(p(x_2); \theta(S_2))}{\partial S_2}}}_{\text{"Stochastic" externality}} \right] \right] Y_{k,1} \right) \right]$$

In this case, the implied corrective tax for the CDF and quantile function formulation is:

$$\tau = \beta \mathbb{E}_{x_1, x_2} \left[\underbrace{\frac{u'(c_2)}{u'(c_1)} D_S(S_2, x_2) Y(k_2)}_{\text{"Standard" externality}} - \underbrace{\frac{u'(c_2)}{u'(c_1)} D_x(S_2, x_2) Y(k_2) \frac{\frac{\partial F(x_2; \theta(S_2))}{\partial S_2}}{\frac{\partial F(x_2; \theta(S_2))}{\partial x_2}}}_{\text{"Stochastic" externality}} \right]$$

$$\tau = \underbrace{\beta \mathbb{E}_{x_1, x_2} \left[\frac{u'(c_2)}{u'(c_1)} D_S(S_2, x_2) Y(k_2) \right]}_{\text{"Standard" externality}} + \underbrace{\beta \mathbb{E}_{x_1, x_2} \left[\frac{u'(c_2)}{u'(c_1)} D_x(S_2, x_2) Y(k_2) \frac{\partial Q(p(x_2); \theta(S_2))}{\partial S_2} \right]}_{\text{"Stochastic" externality}}$$

B Proof of Proposition 2.2.2

The normal PDF is $f(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$. In that case,

$$\frac{\partial \ln(f(\cdot))}{\partial S_2} = \left[\frac{\mu'(S_2)}{\sigma(S_2)} z + (z^2 - 1) \frac{\sigma'(S_2)}{\sigma(S_2)} \right]$$

Here, $z \sim N(0, 1)$. Hence, we can rewrite the "risk endogeneity" term as follows.

$$\mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} \frac{\partial \ln(f(\cdot))}{\partial S_2} \right] = \frac{\mu'(S_2)}{\sigma(S_2)} \mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} z \right] + \frac{\sigma'(S_2)}{\sigma(S_2)} \mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} (z^2 - 1) \right]$$

Because labor is assumed to be exogenous and the shock is at the last period,

$$c_2(x_2) = (1 - D(S_2, x_2))Y(k_2) + (1 - \delta)k_2 \implies \begin{cases} \frac{\partial c_2}{\partial x_2} = -\frac{\partial D(S_2, x_2)}{\partial x_2} Y(k_2) \leq 0 \\ \frac{\partial^2 c_2}{\partial x_2^2} = -\frac{\partial^2 D(S_2, x_2)}{\partial x_2^2} Y(k_2) \leq 0 \end{cases}$$

By Stein's lemma and assuming $u_c \geq 0$ as well as $u_{cc} \leq 0$,

$$\begin{aligned} \mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} z \right] &= \sigma(S_2) \mathbb{E} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial c_2}{\partial x_2} \right] \leq 0 \\ \mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} (z^2 - 1) \right] &= \sigma(S_2)^2 \mathbb{E} \left[\frac{u''(c_2)}{u'(c_1)} \left(\frac{\partial c_2}{\partial x_2} \right)^2 + \frac{u'(c_2)}{u'(c_1)} \frac{\partial^2 c_2}{\partial x_2^2} \right] \end{aligned}$$

Although $\frac{\partial^2 c_1}{\partial x^2}$ could be positive or negative depending on the convexity of damages, we are squaring it and therefore making the entire term positive. Since we assumed $u_{cc} \leq 0$, the first term in the second equation is negative. However, the second term's sign depends on $\frac{\partial^2 c_2}{\partial x_2^2} = -D_{xx}Y(k_2)$, the curvature of damages in x . If we rewrite the "risk endogeneity" term,

$$\begin{aligned} \mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} \frac{\partial \ln(f(\cdot))}{\partial S_2} \right] &= -\mu'(S_2) \mathbb{E} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial D(\cdot)}{\partial x_2} Y(k_2) \right] \\ &\quad + \sigma'(S_2)\sigma(S_2) \mathbb{E} \left[\frac{u''(c_2)}{u'(c_1)} \left(\frac{\partial D(\cdot)}{\partial x_2} Y(k_2) \right)^2 \right] \\ &\quad - \sigma'(S_2)\sigma(S_2) \mathbb{E} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial^2 D(\cdot)}{\partial x_2^2} Y(k_2) \right] \end{aligned}$$

The Pareto PDF is $f(x; \alpha) = \alpha x_m^\alpha x^{-(\alpha+1)}$ for $x \geq x_m$. In that case,

$$\frac{\partial \ln(f(x; \alpha(S_2)))}{\partial S_2} = \frac{\alpha'(S_2)}{\alpha(S_2)} \left[1 - \alpha(S_2) \ln \left(\frac{x}{x_m} \right) \right]$$

Let us define $y \equiv \ln \left(\frac{x}{x_m} \right)$. It is known that $y \sim \exp(\alpha)$, so we have $y \sim \exp(\alpha(S_2))$. Hence,

$$\begin{aligned} \mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} \frac{\partial \ln(f(\cdot))}{\partial S_2} \right] &= \frac{\alpha'(S_2)}{\alpha(S_2)} \mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} - \alpha(S_2) \frac{u(c_2)}{u'(c_1)} y \right] \\ &= \frac{\alpha'(S_2)}{\alpha(S_2)} \mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} \right] - \alpha'(S_2) \mathbb{E} \left[\frac{u(c_2)}{u'(c_1)} y \right] \end{aligned}$$

C Proof of Proposition 3.0.1

The problem (3) has the following Lagrangian.

$$\begin{aligned}\mathcal{L} = & u(c) + \beta \int V(k', S', x') f(x'; \theta(S')) dx' - \lambda [c + k' - (1 - D(S, x))Y(k) - (1 - \delta)k] \\ & - \mu [S' - (1 - \phi)S - Y(k)]\end{aligned}$$

This yields the following first-order conditions.

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial c} : & u'(c) = \lambda \\ \frac{\partial \mathcal{L}}{\partial k'} : & \beta \int \frac{\partial V}{\partial k'} f(x'; \theta(S')) dx' = \lambda \\ \frac{\partial \mathcal{L}}{\partial S'} : & \beta \int \frac{\partial V}{\partial S'} f(x'; \theta(S')) dx' + \beta \int V(k', S', x') \frac{\partial f(x'; \theta(S'))}{\partial \theta} \frac{\partial \theta(S')}{\partial S'} dx' = \mu\end{aligned}$$

By the envelope condition,

$$\frac{\partial V}{\partial S'} = -\lambda' D_S(S', x') Y(k') + (1 - \phi)\mu'$$

If we substitute λ' ,

$$\frac{\partial V}{\partial S'} = -u'(c') D_S(S', x') Y(k') + (1 - \phi)\mu'$$

Substituting into the FOC with respect to S' ,

$$\begin{aligned}\mu = & -\beta \int [u'(c') D_S(S', x') Y(k') - (1 - \phi)\mu'] f(x'; \theta(S')) dx' \\ & + \beta \int V(k', S', x') \frac{\partial f(x'; \theta(S'))}{\partial \theta} \frac{\partial \theta(S')}{\partial S'} dx'\end{aligned}$$

Using $\frac{d \ln(f(x))}{dx} = \frac{f'(x)}{f(x)}$, we can put everything under the same expectation.

$$\mu = -\beta \int \left[u'(c') D_S(S', x') Y(k') - V(k', S', x') \frac{\partial \ln(f(x'; \theta(S')))}{\partial \theta} \frac{\partial \theta(S')}{\partial S'} - (1 - \phi) \mu' \right] f(x'; \theta(S')) dx'$$

By the envelope condition,

$$\begin{aligned} \frac{\partial V}{\partial k'} &= \lambda' (1 - \delta + (1 - D(S', x')) Y'(k')) + \mu' Y'(k') \\ &= \lambda' \left(1 - \delta + \left(1 - D(S', x') + \frac{\mu'}{\lambda'} \right) Y'(k') \right) \end{aligned}$$

Hence, after combining the first two FOCs,

$$\begin{aligned} u'(c) &= \beta \int u'(c') \left(1 - \delta + \left(1 - D(S', x') - \beta \int \left[\frac{u'(c'')}{u'(c')} D_S(S'', x'') Y(k'') \right. \right. \right. \\ &\quad \left. \left. \left. - \frac{V(k'', S'', x'')}{u'(c')} \frac{\partial \ln(f(x''; \theta(S''))}{\partial S''} \right. \right. \right. \\ &\quad \left. \left. \left. - (1 - \phi) \frac{\mu''}{u'(c')} \right] f(x''; \theta(S'')) dx'' \right) Y'(k') \right) f(x'; \theta(S')) dx' \end{aligned}$$

By induction, we can put every $u'(\cdot)$ term and V term together in a weighted infinite sum. Meanwhile, the $(1 - \phi)\mu$ term vanishes by the transversality condition for the condition stock since $\beta(1 - \phi) < 1$ and μ is bounded.

$$\begin{aligned} u'(c_t) &= \beta \int u'(c_{t+1}) \left(1 - \delta + \left(1 - D_{t+1} - \beta \int \left[\sum_{j=0}^{\infty} ((1 - \phi)\beta)^j \frac{u'(c_{t+j+2})}{u'(c_{t+1})} D_S(S_{t+j+2}, x_{t+j+2}) Y(k_{t+j+2}) \right. \right. \right. \\ &\quad \left. \left. \left. - ((1 - \phi)\beta)^j \frac{V_{t+j+2}}{u'(c_{t+1})} \frac{\partial \ln(f(x_{t+j+2}; \theta(S_{t+j+2})))}{\partial S_{t+j+2}} \right] f(\cdot) \right) Y'(k_{t+1}) \right) f(x_{t+1}; \theta(S_{t+1})) dx_{t+1} \end{aligned}$$

Hence, the corrective tax for period t is:

$$\tau_t = \beta \mathbb{E} \left[\sum_{j=0}^{\infty} \beta^j (1 - \phi)^j \left[\frac{u'(c_{t+j+2})}{u'(c_{t+1})} D_{S,t+j+2} Y_{t+j+2} - \frac{V_{t+j+2}}{u'(c_{t+1})} \frac{\partial \ln(f_{t+j+2})}{\partial S_{t+j+2}} \right] \right]$$

In covariance form,

$$\tau_t = \beta \mathbb{E} \left[\sum_{j=0}^{\infty} \beta^j (1 - \phi)^j \frac{u'(c_{t+j+2})}{u'(c_{t+1})} D_{S,t+j+2} Y_{t+j+2} \right] - \sum_{j=0}^{\infty} \beta^j (1 - \phi)^j \text{Cov} \left(\frac{V_{t+j+2}}{u'(c_{t+1})}, \frac{\partial \ln(f_{t+j+2})}{\partial S_{t+j+2}} \right)$$

We can also write the stochastic term in CDF or quantile form. Consider the stochastic part of the tax for a given j :

$$\int V(k_{t+j+2}, S_{t+j+2}, x_{t+j+2}) \frac{\partial \theta(S_{t+j+2})}{\partial S_{t+j+2}} \frac{\partial f(x_{t+j+2}; \theta(S_{t+j+2}))}{\partial \theta} dx_{t+j+2}$$

Integration by parts with $u = V_{t+j+2} \frac{\partial \theta}{\partial S}$ and $dv = \frac{\partial f}{\partial \theta} dx$ yields:

$$\begin{aligned} &= V_{t+j+2} \frac{\partial \theta}{\partial S} \frac{\partial F(x_{t+j+2}; \theta)}{\partial \theta} \Big|_{\underline{x}}^{\bar{x}} \\ &\quad - \int_{\underline{x}}^{\bar{x}} \frac{\partial V_{t+j+2}}{\partial x_{t+j+2}} \frac{\partial \theta}{\partial S} \frac{\partial F(x_{t+j+2}; \theta)}{\partial \theta} dx_{t+j+2} \end{aligned}$$

Here, the first term vanishes by the boundary condition. By the envelope theorem,

$$\frac{\partial V(k, S, x)}{\partial x} = -u'(c) D_x(S, x) Y(k)$$

Therefore,

$$\int V_{t+j+2} \frac{\partial \theta}{\partial S} \frac{\partial f}{\partial \theta} dx_{t+j+2} = \int u'(c_{t+j+2}) D_x(S_{t+j+2}, x_{t+j+2}) Y(k_{t+j+2}) \frac{\partial \theta}{\partial S} \frac{\partial F(x_{t+j+2}; \theta)}{\partial \theta} dx_{t+j+2}$$

Multiplying and dividing by f , the CDF form of the tax is:

$$\tau_t = \beta \mathbb{E} \left[\sum_{j=0}^{\infty} \beta^j (1 - \phi)^j \left[\frac{u'(c_{t+j+2})}{u'(c_{t+1})} D_{S,t+j+2} Y_{t+j+2} - \frac{u'(c_{t+j+2})}{u'(c_{t+1})} D_{x,t+j+2} Y_{t+j+2} \frac{\frac{\partial F(x_{t+j+2}; \theta(S_{t+j+2}))}{\partial S_{t+j+2}}}{\frac{\partial F(x_{t+j+2}; \theta(S_{t+j+2}))}{\partial x_{t+j+2}}} \right] \right]$$

Applying the quantile identity,

$$\frac{\frac{\partial F(x; \theta(S))}{\partial \theta} \frac{\partial \theta(S)}{\partial S}}{f(x; \theta(S))} = - \frac{\partial Q(p; \theta(S))}{\partial \theta} \frac{\partial \theta(S)}{\partial S}$$

Hence, the quantile form of the tax is:

$$\begin{aligned} \tau_t = & \beta \mathbb{E} \left[\sum_{j=0}^{\infty} \beta^j (1 - \phi)^j \frac{u'(c_{t+j+2})}{u'(c_{t+1})} D_{S,t+j+2} Y_{t+j+2} \right] \\ & + \beta \mathbb{E} \left[\sum_{j=0}^{\infty} \beta^j (1 - \phi)^j \frac{u'(c_{t+j+2})}{u'(c_{t+1})} D_{x,t+j+2} Y_{t+j+2} \frac{\partial Q(p(x_{t+j+2}); \theta(S_{t+j+2}))}{\partial S_{t+j+2}} \right] \end{aligned}$$

D DICE climate module

The carbon cycle and climate dynamics follow the standard DICE specification unchanged. The carbon cycle is governed by the linear three-reservoir system:

$$\begin{aligned} M_{\text{AT},t+1} &= b_{11} M_{\text{AT},t} + b_{21} M_{\text{U},t} + E_t + E_t^{\text{land}} \\ M_{\text{U},t+1} &= b_{12} M_{\text{AT},t} + b_{22} M_{\text{U},t} + b_{32} M_{\text{L},t} \\ M_{\text{L},t+1} &= b_{23} M_{\text{U},t} + b_{33} M_{\text{L},t} \end{aligned} \quad (8)$$

where $E_t = \sigma_t(1 - \mu_t)\tilde{Y}_t$ denotes industrial emissions (with σ_t the carbon intensity of output and $\mu_t \in [0, 1]$ the mitigation rate), and E_t^{land} captures exogenous land-use emissions. The climate responds to CO₂ concentrations through the temperature equations:

$$\begin{aligned} T_{\text{AT},t+1} &= T_{\text{AT},t} + c_1 \left[\frac{F_{2 \times \text{CO}_2}}{\ln 2} \ln \left(\frac{M_{\text{AT},t}}{M_{\text{AT}}^{\text{pre}}} \right) + F_t^{\text{EX}} - \lambda T_{\text{AT},t} - c_3(T_{\text{AT},t} - T_{\text{OC},t}) \right] \\ T_{\text{OC},t+1} &= T_{\text{OC},t} + c_4(T_{\text{AT},t} - T_{\text{OC},t}) \end{aligned} \quad (9)$$

where $F_{2 \times \text{CO}_2}$ is the forcing from doubling CO₂, $M_{\text{AT}}^{\text{pre}}$ is the preindustrial atmospheric concentration, F_t^{EX} is exogenous non-CO₂ forcing, λ is the climate sensitivity parameter, and c_1, c_3, c_4 are heat transfer coefficients.

Tauchen discretization

We discretize the AR(1) shock process using $K = 20$ equally spaced grid points on $[-m\sigma_x, m\sigma_x]$ where $m = 3$, giving step size $\Delta = 2m\sigma_x/(K-1)$. The transition probabilities of the resulting Markov chain are:

$$P_{ij} = \Phi \left(\frac{x_j + \Delta/2 - \rho x_i}{\sigma_\varepsilon} \right) - \Phi \left(\frac{x_j - \Delta/2 - \rho x_i}{\sigma_\varepsilon} \right)$$

for interior states $j \in \{2, \dots, K-1\}$, where Φ denotes the standard normal CDF. The boundary states accumulate the tail probabilities: $P_{i1} = \Phi((x_1 + \Delta/2 - \rho x_i)/\sigma_\varepsilon)$ and $P_{iK} = 1 - \Phi((x_K - \Delta/2 - \rho x_i)/\sigma_\varepsilon)$. In the endogenous model, σ_ε is replaced by $\sigma(S_{\text{eff}})$ and ρx_i by $\rho x_i + \mu(S_{\text{eff}})$, yielding the state-dependent matrix $P_{ij}(S)$ of equation (10).

D.1 State-dependent Tauchen discretization

The Tauchen grid is constructed once, wide enough to accommodate the worst-case distribution at the upper bound $S_{\text{eff}}^{\max} = 5.0^\circ\text{C}$. The grid points $\{x_1, \dots, x_K\}$ remain fixed, but the transition matrix is recomputed at each period based on the realized effective pollution state:

$$P_{ij}(S) = \Phi\left(\frac{x_j + \Delta/2 - \rho x_i - \mu(S)}{\sigma(S)}\right) - \Phi\left(\frac{x_j - \Delta/2 - \rho x_i - \mu(S)}{\sigma(S)}\right) \quad (10)$$

This is the discrete analogue of the theoretical conditional density $f(x'; \theta(S'))$ where the parameter vector $\theta = (\mu, \sigma)$ varies with the pollution stock. Unlike the exogenous case where P is constant over time, $P(S_{\text{eff},t})$ evolves with the climate trajectory, making the Markov chain *non-stationary*.